

MD RAKIBUL HASAN

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U.S. Permanent Resident – authorized to work without sponsorship

Summary

Machine learning professional with 7+ years of experience building predictive models and data-driven optimization solutions on 12TB+ datasets across a 53M-subscriber network. Developed and applied ML methods for churn prediction, time-series forecasting, customer segmentation, and campaign optimization, driving 11% regional revenue growth and 16% improvement in campaign efficiency. PhD candidate in Computer Science at NJIT, with 4 peer-reviewed publications on reinforcement learning and adaptive optimization, and a track record of translating large-scale data into deployable ML solutions with measurable business impact.

Tech stack: Python, SQL, Scikit-learn, PyTorch, Hive/Hadoop/Spark, Databricks, Docker, Power BI, Git.

Education

PhD in Computer Science, New Jersey Institute of Technology (NJIT), USA

December 2026

GPA: 3.91/4.0

MBA, Institute of Business Administration, University of Dhaka, Bangladesh

2021

GPA: 3.82/4.0 (Valedictorian)

BS in Electrical & Electronic Engineering, BUET, Bangladesh

2013

Experience

PhD Researcher (ML, Database)

09/2021 – Present

NJIT, Newark, NJ, USA

- Reduced SQL query execution time by **30%** on dynamic workloads by developing a **Dueling NoisyNet-DQN** tuner for adaptive database index selection, outperforming existing learning-based baselines.
- Lowered total query time and index transition overhead by building a **Bayesian Q-learning framework** for online index tuning, achieving **21× less transition cost** than Multi-Armed Bandit and RL-based advisors via Thompson Sampling.
- Accelerated graph pattern query evaluation by **2.7×** by designing a **summary graph-based materialized view** framework that prunes search space while reducing storage requirements on large real-world graphs.

Manager (ML & DS), Telecom Market Strategy & Planning

10/2020 – 08/2021

Robi Axiata (53M+ subscribers), Bangladesh

- Improved regional campaign efficiency by **16%** by building **segmentation models** across geographic, demographic, revenue, and cost dimensions to prioritize market investments.
- Developed **dynamic time-series forecasting** models for revenue chains achieving **95% forecast accuracy**, enabling strategic investment decisions at regional and national levels.
- Conducted **customer churn analysis** combining CDP data and network traffic insights; recommended targeted retention strategies that delivered an **11% regional revenue boost**.
- Designed and deployed an **ML-driven marketing campaign** for OTT platform (Binge), increasing new user acquisitions by **21%** and improving campaign efficiency by **16%** through personalized promotions.
- Led monthly/quarterly performance reviews with variance analysis, resulting in **12% improvement in regional KPIs**.

Specialist (ML), Core Network Planning

01/2019 – 09/2020

Robi Axiata Limited, Bangladesh

- Optimized core network resource allocation by building a **neural network model** to identify parameters most impacting RNC/BSC (network controller) processor load, triggering data-driven site rehomings that reduced **RNC utilization by 15%**.
- Prevented **12% daily 3G traffic loss** during COVID-19 by forecasting pandemic-driven user mobility with **LSTM and HMM** on historical time-series and marketing data.
- Led virtualization of **11 MSCs and 6 SGSNs** (core network switches) within 5 months; migrated **3 legacy HSS databases** (subscriber registry, **66M users**) to virtual infrastructure with <100 customer complaints.

Specialist (ML), Radio Network Planning

09/2016 – 12/2018

Robi Axiata Limited, Bangladesh

- Trained **clustering models** to identify 7,000 high-potential sites for LTE rollout, optimizing 4.5G coverage prioritization across 64 districts of Bangladesh.
- Prevented **8% daily 4G data loss** by deploying **LSTM/HMM forecasting** models for traffic trend prediction, enabling proactive capacity planning.
- Led RF integration of **4,500 sites** during the Robi-Airtel merger (largest telecom merger in Bangladesh), managing **23 engineers** and reducing customer complaints by **29%** within 3 days of site rehomings.
- Achieved **62% capacity increase** through dynamic spectrum sharing optimization and frequency refarming strategies.

- Optimized new customer acquisition using **audience segmentation and logistic regression**, adding **50,000 new data customers** in a single quarter.

Research Intern (ML & DS)

09/2019 – 12/2019

Fujitsu Research Institute, Tokyo, Japan

- Improved autonomous vehicle braking system response by analyzing time-series car sensor data using **LSTM**, reducing false-positive braking events in simulation testing.
- Increased convenience-store daily sales by **2.1%** and customer engagement by **3.6%** by mining POS data with Gradient Boosting, Eclat, and Apriori algorithms.
- Predicted automotive manufacturer monthly demand at **95% confidence** using LSTM, SARIMA, and Prophet on historical sales and macroeconomic data.

RF Optimization Engineer

01/2014 – 08/2016

Huawei Technologies, Bangladesh

- Enabled real-time monitoring of **3,800+ NodeBs** by implementing **enhanced Markov algorithm** for user mobility prediction and **MapReduce** pipelines for network KPI/KQI analytics on a Hadoop big data platform.
- Automated targeted promotional campaigns across millions of subscribers by developing **scalable ML classification models** on Hadoop to segment customer interests, replacing manual campaign selection.
- Transformed managed-service projects from monthly penalty to reward by leading **7 optimization teams** covering **1,450+ NodeBs** and optimizing KPIs (RRC failure, RAB assignment, CS/PS drop, congestion).

Projects

- **Database Q&A Chatbot (RAG)**: Built a retrieval-augmented (RAG) Q&A chatbot over a 631-page technical corpus, delivering source-cited, hallucination-free answers on 5 of 6 benchmark questions by combining ChromaDB semantic retrieval, a vision LLM that recovered 100+ diagram/math slides (~16% of the corpus), and a 6-clause grounding prompt at temperature 0.1.
- **Customer Lifetime Value Model**: Built probabilistic CLV prediction using BG/NBD and Gamma-Gamma models for marketing budget allocation.
- **Network Anomaly Detection**: Developed unsupervised ML system using Isolation Forest to detect traffic anomalies in real-time.
- **LLM Fine-Tuning Pipeline**: Fine-tuned Llama-2 7B for text-to-SQL generation using LoRA and QLoRA with Hugging Face PEFT. Deployed inference endpoint on AWS SageMaker with 4-bit quantization.

Technical Skills

- **Languages**: Python, C++, SQL, MATLAB
- **ML & Data Science (DS)**: Scikit-learn, PyTorch, Pandas, NumPy, Matplotlib, SciPy; Regression, Classification, Clustering, Time-Series Forecasting, Anomaly Detection, Reinforcement Learning, Deep Learning (LSTM, CNN, DQN), GenAI, LLM, Agentic AI
- **Big Data & Databases**: Hive SQL, Hadoop/HDFS, Apache Spark, Databricks, MySQL, PostgreSQL, IBM Db2
- **Visualization & BI**: Power BI, Tableau, Excel VBA
- **Tools & Methods**: Git, SPSS, SAS, Statistical Modeling, A/B Testing, ETL Pipelines, Feature engineering pipeline, Batch inference, Docker

Publications

- “Multi-Objective Reinforcement Learning for Index Selection in Hybrid Transactional/Analytical Workloads” — **ICDE (In review)**
- “Database Tuning via Distributional Reinforcement Learning for Hybrid Transactional/Analytical Processing Workloads” — **DaWaK 2026**
- “Adaptive Online Index Selection with a Noisy Dueling Deep Q-Network” — **WISE 2025**
- “A Bayesian Reinforcement Learning Framework for Online Index Tuning” — **DaWaK 2025**
- “Scalable Optimization of Graph Pattern Queries Using Summary Graphs” — **WISE 2024**
- “LiteSelect: A Lightweight Adaptive Learning Algorithm for Online Index Selection” — **DaWaK 2024**
- “Performance of successive interference cancellation technique in IEEE 802.11 based WLANs” — **ICIEV 2013**

Certifications

- Scalable Machine Learning on Big Data using **Apache Spark** — Coursera, 2020
- Data Science on **Google Cloud Platform**: Predictive Analytics — Coursera, 2020
- **IBM AI Engineering Specialization** — Coursera

Awards & Recognition

- **Gold Medal & Valedictorian**, Executive MBA, IBA, University of Dhaka (highest CGPA, 2021)

- **Merger Hero 2017**, Robi Axiata — for leading RF integration during Robi-Airtel merger preserving baseline KPIs
- **Q2'2019 Technology Champion**, Robi Axiata — for managing network capacity during festival traffic surge